

AI-Powered Pension Knowledge Search System Architecture

1. Overview

This document describes the architecture of an AI-powered system designed to help customer support staff retrieve accurate pension-related articles using natural language queries. The system uses Retrieval-Augmented Generation (RAG) to provide relevant answers and article links.

2. Objectives

- Enable semantic search (not keyword-based)
- Reduce training time for new staff
- Provide accurate answers with source articles
- Improve customer support efficiency

3. High-Level Architecture

User -> API Layer -> Embedding Model -> Vector Database -> Retrieval -> LLM -> Response

4. Components

4.1 User Interface:

Web or internal tool where staff ask questions.

4.2 API Layer (FastAPI):

Handles requests and responses.

4.3 Embedding Model:

Converts text into vector representations.

4.4 Vector Database:

Stores embeddings for fast similarity search (ChromaDB/FAISS).

4.5 Retrieval Engine:

Finds closest matching article chunks.

4.6 LLM (Optional):

Generates human-readable answers.

4.7 Article Repository:
Source of all pension-related articles.

5. Data Flow

1. User enters a question
2. Question is converted to embedding
3. Vector DB retrieves similar content
4. Relevant articles are returned
5. LLM generates response (optional)
6. Response + article links displayed

6. Data Processing Pipeline

1. Collect articles
2. Clean text
3. Chunk documents
4. Generate embeddings
5. Store in vector DB

7. Technology Stack

Backend: FastAPI

Embeddings: SentenceTransformers/OpenAI

Vector DB: ChromaDB/FAISS

LLM: OpenAI/Azure

Database: PostgreSQL

Deployment: Docker/Cloud

8. Security and Compliance

Ensure responses are based only on verified articles. Avoid hallucinations. Log queries and responses.

9. Future Enhancements

Admin dashboard, analytics, feedback loop, multilingual support, auto-updates.